

Ionic bonding

Learning outcomes

By the end of this section you should be able to:

- Define ionic bonding.
- Deduce the formula and name of an ionic compound from its component ions.
- Interconvert names and formulas of binary ionic compounds.

It is important to know what the charge of ions are within a compound. Some species, such as transition elements, can form ions with different charges. The different charged ions can have completely different properties even though they are the same element!

Fehling's reagent is an example of a solution containing copper ions that is used to detect glucose in urine as a tool to screen for diabetes. The original solution is a blue, homogeneous mixture containing Cu^{2+} ions. When the solution comes in contact with glucose a reaction occurs converting the Cu^{2+} ions to Cu^+ ions. These Cu^+ ions form Cu_2O and create a heterogeneous mixture indicating the presence of glucose in solution. This process is shown in **Figure 1**.

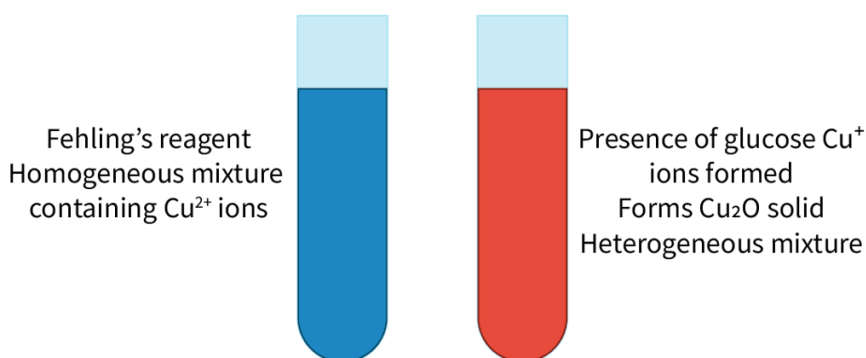


Figure 1. Fehling's reagent is used to detect the presence of glucose.

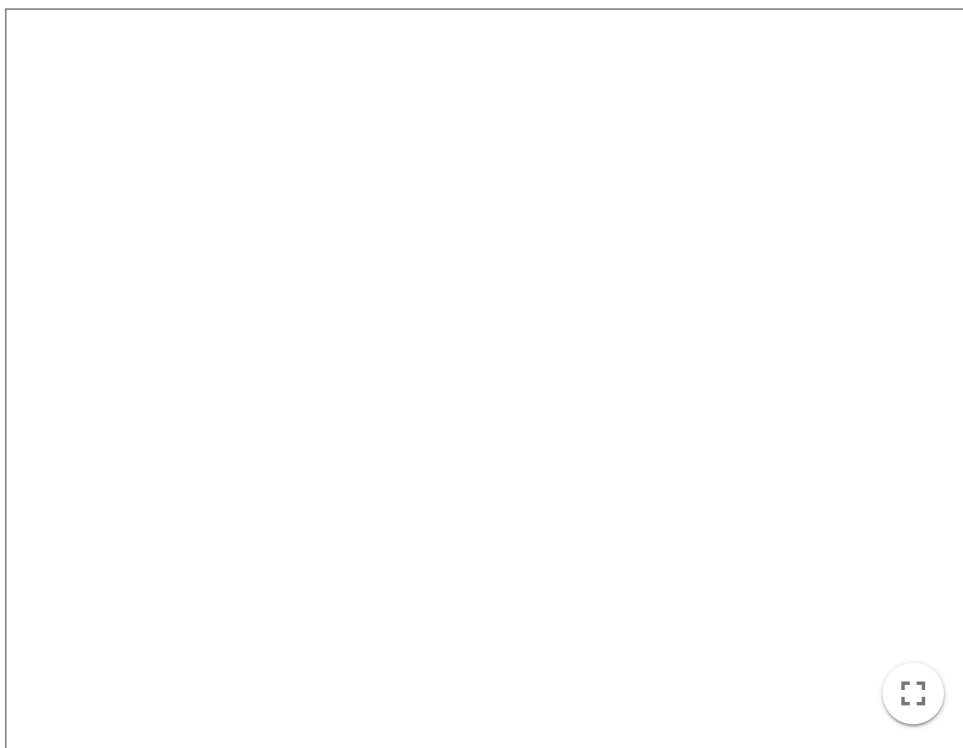
Another example of ions with varying charges is chromium. Chromium can exist as different ions such as Cr^{3+} and Cr^{6+} . In small quantities, Cr^{3+} is an essential nutrient found in foods that is required for sugar and fat metabolism in mammals. Cr^{6+} , however, is toxic and carcinogenic. Industrial pollution has caused contamination of some groundwater supplies with Cr^{6+} globally. One well known

example of Cr^{6+} contamination was the Hinkley groundwater contamination made famous by the film *Erin Brockovich*. You can read more about this example [here](https://www.scientificamerican.com/article/chromium-water-cancer/)  (<https://www.scientificamerican.com/article/chromium-water-cancer/>).

What happens when a copper or chromium cation encounters an anion? How can we look at a chemical formula, such as Cu_2O or CrO_3 and know whether the chromium ion within it is the one that acts as an essential nutrient or the toxic one?

Recall from [section S2.1.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>) that when metal atoms lose electrons they form positive ions called cations and when non-metal atoms gain electrons they form negative ions called anions. What do you think happens if a metal is in the presence of a non-metal? The metal element can lose electrons and the non-metal element can gain the lost electrons. This process of electrons being transferred from one atom and gained in another happens simultaneously and forms ions of opposite charge. What do you think happens when ions of opposite charge are in close proximity to one another? Oppositely charged ions experience an electrostatic attraction. This is what occurs in the formation of an ionic bond and is illustrated in **Interactive 1**.

Press the 'next' button to watch the formation of an ionic bond.



Interactive 1. Formation of the ionic bond.

 [More information for interactive 1](#)

An interactive slideshow with five slides demonstrates the ionic bonding between Sodium and Chlorine. Next and Previous buttons are provided on the bottom, allowing users to navigate between the slides.

Slide 1: An outline of the periodic table with highlighted Sodium (metal) on the left side and Chlorine (non-metal) on the right side. Sodium is in Group 1 and Period 3. Chlorine is in Group 17 and Period 3.

Slide 2: Sodium is depicted as a large circle and chlorine as a small circle. The electronic configuration of sodium is $[\text{Ne}] 3s^1$. The electronic configuration of chlorine is $[\text{Ne}] 3s^2 3p^5$. Sodium loses one electron to have a full valence shell, and Chlorine gains one electron to have a full valence shell.

Slide 3: An animation explains how sodium transfers its valence electron to chlorine. One electron is transferred from sodium to chlorine. Ions are formed after the transfer of electrons occurs. Sodium becomes a cation (Na^+) and is represented as a small circle, while chlorine becomes an anion (Cl^-) and is represented as a large circle.

Slide 4: An animation illustrates how these oppositely charged ions experience an electrostatic attraction to one another. The Na^+ ion is attracted to the Cl^- ion. The electrostatic attraction between the oppositely charged ions is called the ionic bond.

Slide 5: The Na^+ ion interacts with the Cl^- ion.

A play/pause button is available on the bottom left to control the animations. An option for full screen is available at the bottom right.

This interactivity helps users understand the formation of NaCl through ionic bonding and the electrostatic attraction between sodium and chloride ions.

The ionic bond is formed by the electrostatic attraction between oppositely charged ions. While the individual ions are charged, the ionic compound itself is electrically neutral. For example, sodium chloride is made up of Na^+ and Cl^- ions but the compound is electrically neutral.

Writing the chemical formula for binary ionic compounds

In section S1.4.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/empirical-formula-and-molecular-formula-id-43572/>) you learned about the empirical formula of a compound as the simplest ratio of atoms of each element present in that compound. The chemical formula for ionic compounds is known as a formula unit. It represents the simplest ratio of ions making up the ionic compound and is also an example of an empirical formula. If you look at the formula unit of an ionic compound such as NaCl , this means that there is a 1:1 ratio of Na^+ to Cl^- ions. This does not mean that there is only one of each ion, since ionic compounds are found in a crystal lattice structure consisting of many ions.

You can deduce the formula for binary ionic compounds by considering the number of valence electrons. As ionic compounds are electrically neutral, the ratio of cations to anions is a result of the number of electrons lost from the metal and gained in the non-metal. Let's look at **Figure 2** to illustrate how to deduce the chemical formula for two different ionic compounds.

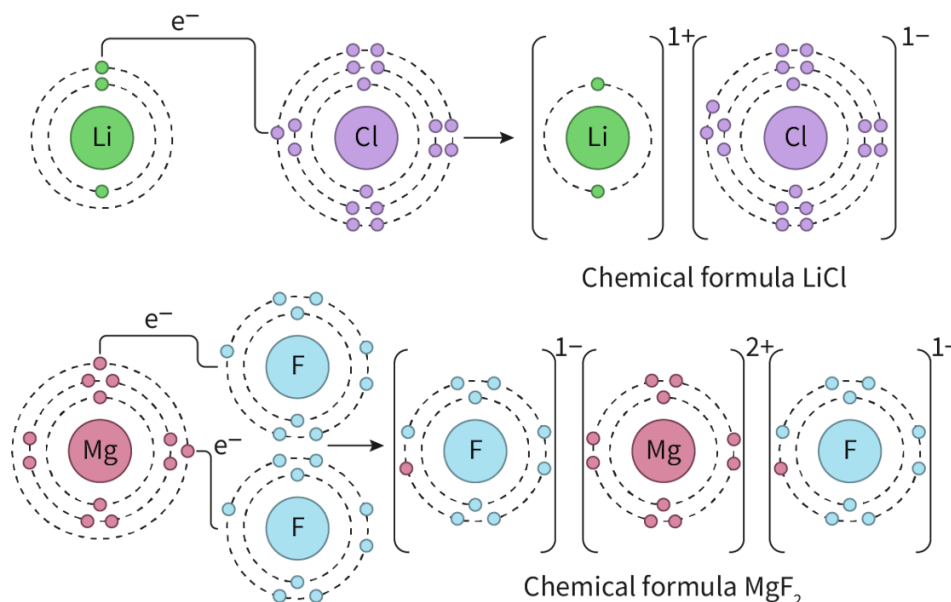


Figure 2. Formation of ionic compounds LiCl and MgF₂.

 More information for figure 2

The image is a diagram illustrating the formation of two ionic compounds: LiCl and MgF₂. In the formation of LiCl, lithium (Li) is shown with one valence electron, while chlorine (Cl) is shown with seven valence electrons. Lithium donates its one electron to chlorine, forming a Li⁺ cation and a Cl⁻ anion, which combine to form LiCl.

In the formation of MgF₂, magnesium (Mg) is illustrated with two valence electrons, while each fluorine atom (F) is shown with seven valence electrons. Magnesium donates one electron to each fluorine atom, resulting in an Mg²⁺ cation and two F⁻ anions. They combine in a 1:2 ratio to form MgF₂. The diagram uses Lewis dot structures to represent the electron transfers and resulting ions.

[Generated by AI]

In the first example in **Figure 2**, lithium has one valence electron so it will form the Li⁺ ion. It can transfer that one valence electron to a chlorine atom to form the Cl⁻ ion. Since they both have the same (but opposite) charge, the ratio of cations to anions will exist in a 1:1 ratio. Therefore, the chemical formula will be LiCl.

Study skills

Note that when writing the chemical formula for ionic compounds the cation is **always** written first. When you are deducing the chemical formula of an ionic compound, it is necessary to know the charges of the individual ion. These charges are helpful to determine the chemical formula, however, ensure that no charges are shown when writing the chemical formula of ionic compounds. The ionic compound is electrically neutral; it is just composed of charged ions.

In the second example in **Figure 2**, magnesium has two valence electrons so will form the Mg^{2+} ion. It has two electrons available for transfer, however, fluorine only needs to gain one electron from magnesium to have a full valence shell. This means there is another electron available to be transferred to a second fluorine atom. Therefore, there are two F^- ions for every one Mg^{2+} ions. The ratio of cations to anions will exist in a 1:2 ratio. Hence, the chemical formula will be MgF_2 .

Making connections

The formation of an ionic compound from its elements, where some elements lose electrons and the others gain electrons, is an example of a redox reaction. You will explore this type of reaction in subtopic R3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/>).

Another way to deduce the formula is to use what is known as a ‘swap and drop’ approach. Using this method, you would:

- deduce the charge on each ion
- if the charges of the ions are different **swap** the number of charges and **drop** the signs
- the simplest ratio of the swapped charges helps you deduce the chemical formula.

Consider the three examples shown in **Table 1**.

Table 1. Swap and drop method for determining chemical formula of ionic compounds

	Sodium fluoride		Sodium sulfide		Magnesium oxide	
Symbol	Na^+	F^-	Na^+	S^{2-}	Mg^{2+}	O^{2-}
Charge	1+	1-	1+	2-	2+	2-
Swap and drop	Charges are the same Write in 1:1 ratio		Charges are different Swap and drop		Charges are the same 2:2 ratio which can be reduced to 1:1 ratio	
Ratio	1	1	2	1	1	1
Chemical formula	NaF		Na_2S		MgO	

	Sodium fluoride	Sodium sulfide	Magnesium oxide
Note	When no subscript is written it is assumed to have a value of one		

Naming binary ionic compounds

You have probably heard of some names of ionic compounds in your everyday life. Sodium chloride in table salt, potassium iodide in iodised salt and sodium fluoride in some fluoride containing toothpastes. Do you notice anything about these names? Which ion is named first? Which ion is named second? Do the names of the ion differ from the names of the neutral element?

Nature of Science

Aspect: Science as a shared endeavour

Scientists aim to develop common terminology and conventions, such as naming ionic compounds, to facilitate unambiguous communications globally.

When naming binary ionic compounds (those made up of only two elements), the cation is listed first, followed by the anion. The name of the cation is simply the same as the neutral element. For the anion, the root of the full name is used, but the suffix is changed to ‘-ide’: for example, when a chlorine atom gains an electron it becomes a **chloride** ion. The subscript does not play a role in the name of an ionic compound since the compound has a fixed ratio of ions to ensure it is electrically neutral.

The name of the cation always remains the same as the name of the element, but the name of the anion changes to have an ‘-ide’ ending. **Table 2** lists some elements and names of anions.

Table 2. Elements and anion names.

Element name	Anion name	Element name	Anion name
Hydrogen	Hydride	Oxygen	Oxide
Fluorine	Fluoride	Sulfur	Sulfide
Chlorine	Chloride	Selenium	Selenide
Bromine	Bromide	Nitrogen	Nitride

Element name	Anion name	Element name	Anion name
Iodine	Iodide	Phosphorus	Phosphic

If the cation is a transition element, roman numerals are used in brackets to denote the charge of the cation when naming. Let's look at a few examples in **Table 3**.

Table 3. Name and chemical formula for a few ionic compounds.

Ions	Name	Chemical formula
Li^+ and S^{2-}	Lithium sulfide	Li_2S
Cu^{2+} and Cl^-	Copper(II) chloride	CuCl_2
Fe^{3+} and O^{2-}	Iron(III) oxide	Fe_2O_3



Exercises 1 – 4

Click a question to answer



1. What is the chemical formula for magnesium iodide?



Type here



2. What is the name for CuO ?



Type here



3. What is the chemical formula for chromium(III) oxide?



Type here



4. What is the name for Na_3N ?



Check answers

Polyatomic ions

Did you know that it is possible to form an ion from more than one atom? Ions can be made up of more than one atom that, overall, has experienced a loss or gain of electrons. These are known as polyatomic ions. These types of ion are made up by atoms held together by covalent bonds but, overall, have experienced a loss or gain of electrons. You will learn more about covalent bonding in section S2.2.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>). This causes this group of atoms to have a positive or negative charge and can form an ionic bond with an ion of opposite charge.

Table 4 shows some of the most common polyatomic ions that you will encounter in this course.

Table 4. Common examples of polyatomic ions.

Name of polyatomic ion	Overall charge of ion	Formula
Ammonium	1+	NH_4^+
Hydroxide	1-	OH^-
Nitrate	1-	NO_3^-
Hydrogencarbonate	1-	HCO_3^-
Carbonate	2-	CO_3^{2-}
Sulfate	2-	SO_4^{2-}
Phosphate	3-	PO_4^{3-}

Study skills

Note that the names and formula for polyatomic ions **are not** provided in the DP chemistry data booklet. This means that you must be familiar with those listed in **Table 4** for assessments.

Figure 3 depicts an ionic bond between the ammonium and chloride ions illustrating that this compound would contain both ionic and covalent bonds.

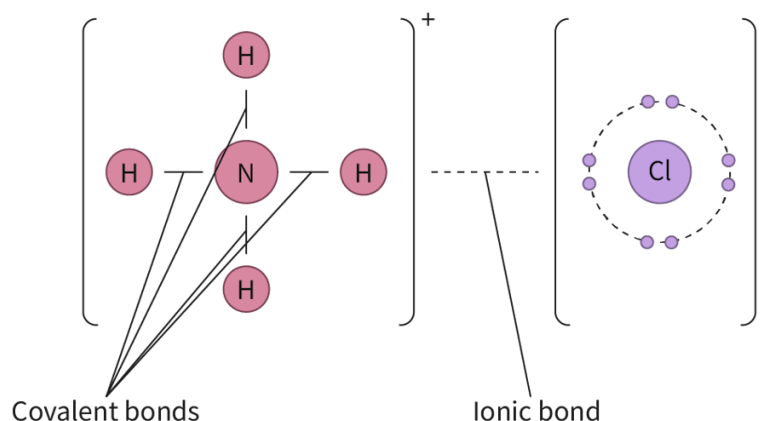


Figure 3. Bonding within ammonium chloride.

 More information for figure 3

The diagram illustrates the bonding structure within ammonium chloride. On the left side, there is an ammonium ion (NH_4^+), depicted as a central nitrogen (N) atom bonded to four surrounding hydrogen (H) atoms, each connected by covalent bonds, shown as lines between the atoms. The entire ammonium ion is enclosed within a bracket marked with a positive sign to indicate its overall positive charge.

On the right side, the chloride ion (Cl^-) is shown, represented by a central chlorine (Cl) atom surrounded by dotted lines depicting a shell with seven electrons, symbolizing its negative charge. The chloride ion is also enclosed within brackets marked with a negative sign.

A dotted line between the ammonium and chloride ions signifies the ionic bond that holds these two parts together, illustrating the combination of ionic and covalent bonding within ammonium chloride.

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How might the names and formulas of ionic compounds containing polyatomic ions differ from those of binary compounds? When writing the chemical formula, treat the entire polyatomic ion as one unit, using brackets when there is more than one of the same polyatomic ion. When naming an ionic compound containing a polyatomic ion, simply state the name of the polyatomic ion. **Figure 4** shows how you can deduce the chemical formula for the ionic compound ammonium phosphate.

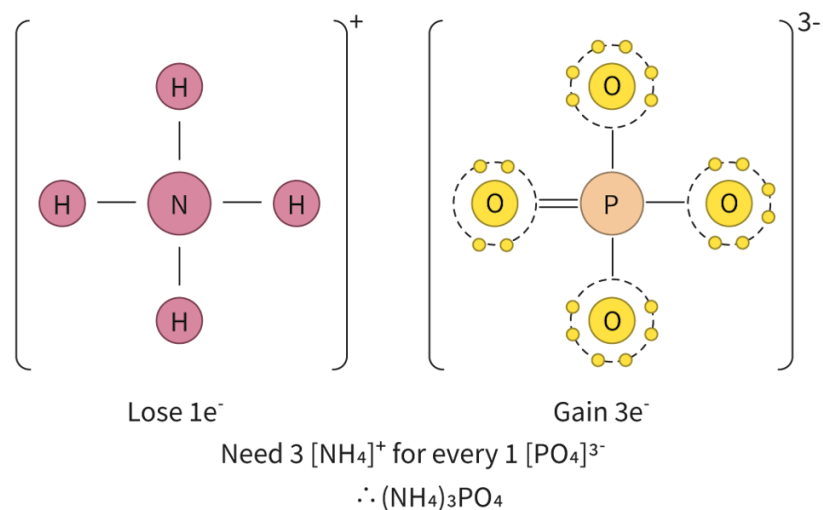


Figure 4. Deducing the chemical formula for ammonium phosphate.

More information for figure 4

The image is a diagram illustrating the chemical relationship between ammonium and phosphate ions to form ammonium phosphate. On the left side, there is a representation of the ammonium ion (NH_4^+), which consists of a central nitrogen (N) atom surrounded by four hydrogen (H) atoms, each connected with a single line, indicating chemical bonds. There is a positive charge symbol near the brackets surrounding the ion, indicating a loss of one electron.

On the right side, the phosphate ion (PO_4^{3-}) is shown with a central phosphorus (P) atom connected to four oxygen (O) atoms with double bonds and enclosed in brackets. Around each oxygen atom, there are dots representing gained electrons, and a negative charge of three is shown outside the brackets.

Below these representations, text reads "Lose $1e^-$ " near the ammonium ion and "Gain $3e^-$ " near the phosphate ion. Further text states "Need 3 $[\text{NH}_4]^+$ for every 1 $[\text{PO}_4]^{3-}$ $\therefore (\text{NH}_4)_3\text{PO}_4$," indicating the combination of ions to form ammonium phosphate.

[Generated by AI]

We can similarly use the swap and drop method to determine the chemical formula for ionic compounds containing polyatomic ions. Using this method, you would:

- deduce the charge on each ion
- if the charges of the ions are different **swap** the number of charges and **drop** the signs
- the simplest ratio of the swapped charges helps you deduce the chemical formula
- if more than one polyatomic ion is needed in the chemical formula, use brackets around it.

Table 5. Swap and drop method for determining chemical formula of ionic compounds involving polyatomic ions.

	Ammonium chloride		Iron(III) sulfate	
Symbol	NH_4^+	Cl^-	Fe^{3+}	SO_4^{2-}
Charge	1+	1-	3+	2-
Swap and drop	Charges are the same Write in 1:1 ratio		Charges are different Swap and drop	
Ratio	1	1	2	3
Chemical formula	NH_4Cl		$\text{Fe}_2(\text{SO}_4)_3$	



Exercise 5

Click a question to answer



1. What is the chemical formula for magnesium iodide?



Check answers

In the next activity you will use a matching puzzle to reinforce your learning about ionic compounds.

Activity

- **IB Learner profile attribute:** Communicators
- **Approaches to Learning:** Social — Collaborating with a peer to consolidate understanding and to work towards a common goal
- **Time required to complete activity:** 15—20 minutes
- **Activity type:** Group activity (2—3 students)

Download the following Tarsia puzzle.

Tarsia puzzle 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry)

In groups of 2–3, complete the Tarsia puzzle by placing the sides of the triangles that match one another beside each other. Ensure all members of the group make equal contributions.